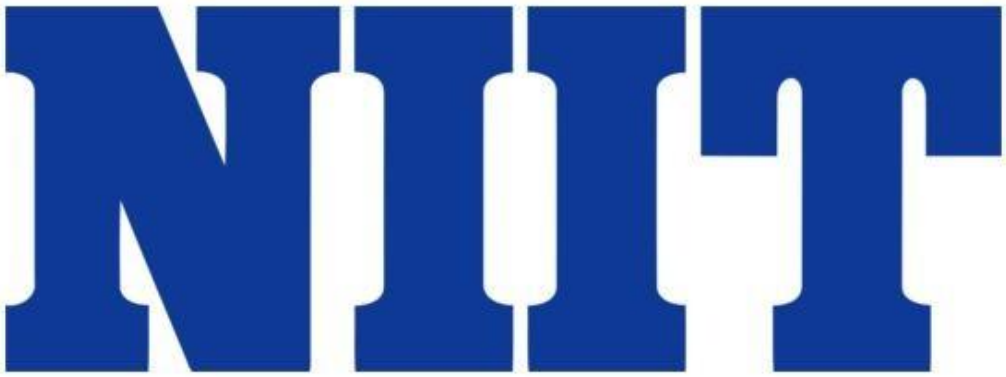


AIIT INTEGRATED TECHNOLOGIES LTD.
DIGITAL FORENSICS & INCIDENT RESPONSE UNIT DIGITAL FORENSIC INVESTIGATION PROJECT

Case ID: AIT-DF-2026-009
Investigation Type: Digital Forensics /Deleted Data Recovery
Investigator: Essel Desire
Team:NIIT COHORTS
Date: 27 September 2026

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CASE BACKGROUND

Apex Integrated Technologies Ltd. Assigned a Digital Forensic Investigation Team to examine a forensic image obtained from a workstation involved in a suspected data-delection incident. During a routine review,the organization discovered that several folders and files that were previously present on the workstation were no longer visible.

Management does not want assumptions.The investigation must rely only on forensic evidence to determine:

- What information was present on the Workstation
- What information was deleted
- What can be recovered
- When the relevant activity occurred
- What evidence can establish the sequence of events
- What can and cannot be concluded from the available evidence

The organization supplied a forensic image of the workstation for examination.

INVESTIGATION OBJECTIVES

The primary objective of this investigation is to examine the supplied forensic image and investigate the disappearance of the following folders:

1. SSH
2. My pic
3. Our pic
4. ADDS Project
5. Corel
6. Corel Draw Work

The Investigation must determine whether these folders or their contents are:

- Present
- Deleted
- Partially deleted
- Recoverable
- Partially recoverable
- Unrecoverable

In addition,the investigation aims to establish the most defensible timeframe in which the deletion activity occurred,based solely on the available forensic evidence.

SCOPE

This investigation is limited to the forensic examination of the supplied E01 disk image.The scope includes identification of the six target folders (SSH,My pic,Our pic,ADDS Project,Corel and Corel Draw

Work),determination of their status (present,deleted or recoverable),recovery of the deleted data where possible,and establishment of a defensible timeline of activity.
Live system analysis,network traffic examination and interviews were outside the scope of this investigation.

EVIDENCE DESCRIPTION

Evidence Information	**Result**
Evidence Filename	Evidence.E01
Evidence Size	9.86 MB(10,341,301 bytes)
Image Format	E01
Partitions(s)	Disk Image
File System(s)	NTFS
Operating System	Windows
User account(s)	Not identified
Acquisition date/time	27 September 27, 2026/ 5:30 pm
SHA-256	44A8BA2C0B1FE943060E869B831EB6CA3781DE7DA8FA3B827953B3C4D00C22F1
MD5	AC9F3AEAD6112049279957D6D726EEB3

EVIDENCE INTEGITY

The integrity of the forensic image was verified using cryptographic hashing.
Hash Algorithms Used: SHA-256 and MD5

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\System32> Get-FileHash -Algorithm SHA256 "C:\Users\ruhuo\Downloads\Evidence.E01"

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          44A8BA2C0B1FE943060E869B831EB6CA3781DE7DA8FA3B827953B3C4D00C22F1       C:\Users\ruhuo\Downloads\Evid...


PS C:\WINDOWS\System32> Get-FileHash -Algorithm MD5 "C:\Users\ruhuo\Downloads\Evidence.E01"

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
MD5             AC9F3AEAD6112049279957D6D726EEB3                                       C:\Users\ruhuo\Downloads\Evid...
```

A Working copy of the image was used for analysis. The hash values of the working copy matched the original hashes,confirming that the evidence was not altered.
Hashing is critical in digital forensics because it provides a unique digital fingerprint of the evidence. If the hash value changes,it indicates that the file has been modified.
If the hash had changed unexpectedly,the integrity of the evidence would be compromised and the findings could be challenged in the court or by another examiner.

TOOLS USED

1. DD: Used to create/examine a forensic disk image and verify evidence.

2. R-DRIVE Image: Used for forensic disk imaging and examination.
3. FTK Imager: Used to examine the forensic disk image,file system,folders,files and deleted data.
4. Autopsy: Used to analyze the forensic image and investigate files,deleted data and artifacts.



About



Autopsy®

OPEN | EXTENSIBLE | FAST

Autopsy™ is a digital forensics platform based on The Sleuth Kit™ and other tools.

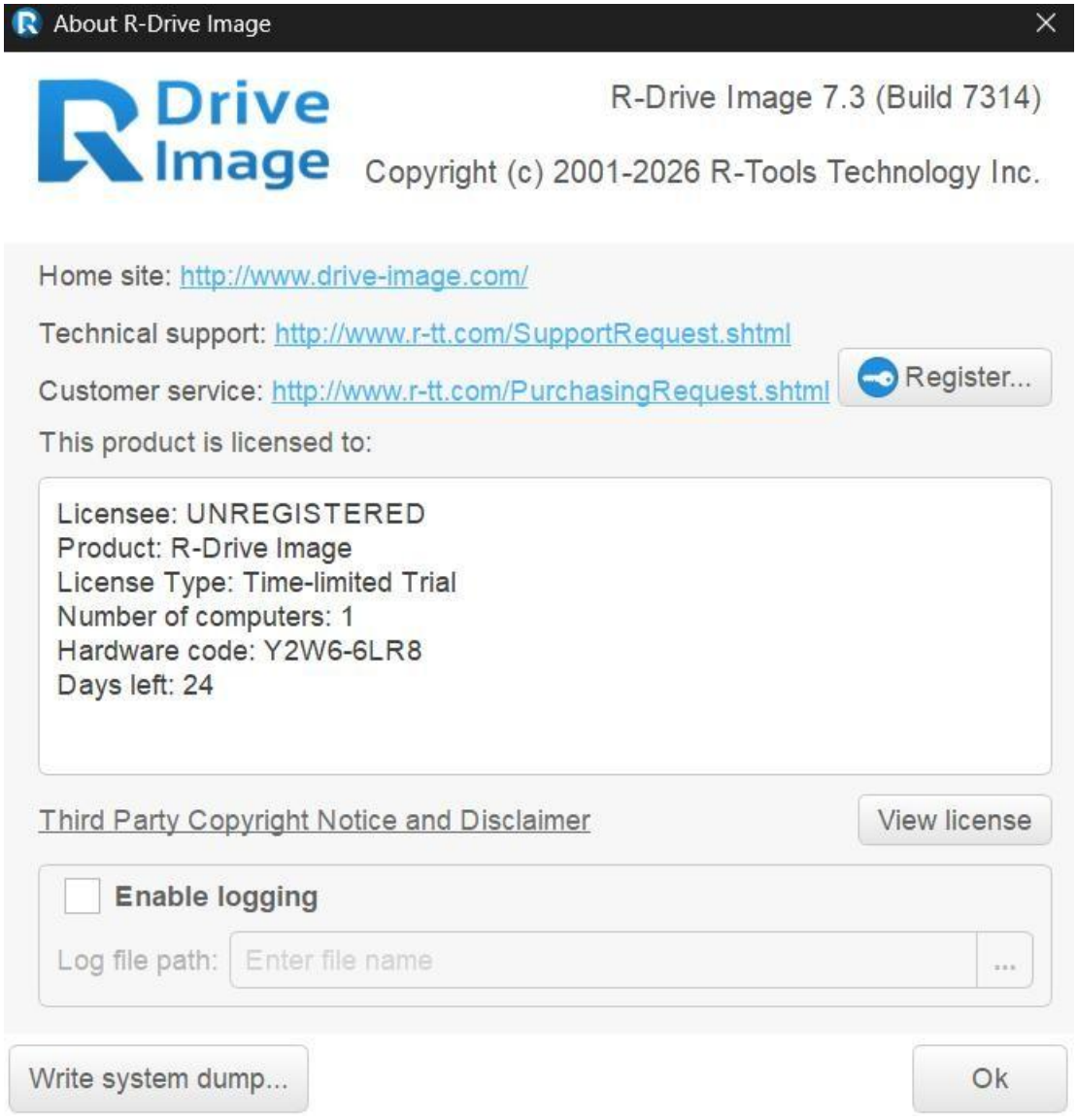
- General Information: http://www.autopsy.com.
- Training: https://www.autopsy.com/support/training/
- Support: https://www.autopsy.com/community/

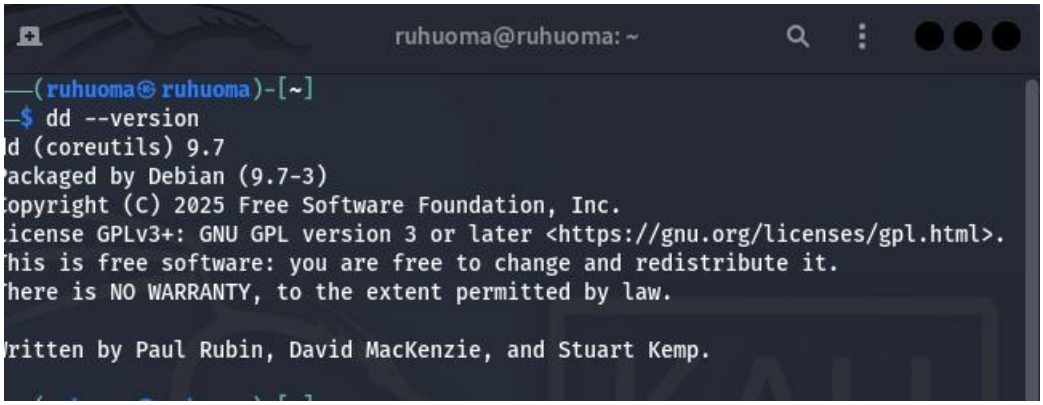
Copyright © 2003-2025 Sleuth Kit Labs, Inc.

Maintained by: Sleuth Kit Labs https://www.sleuthkitlabs.com

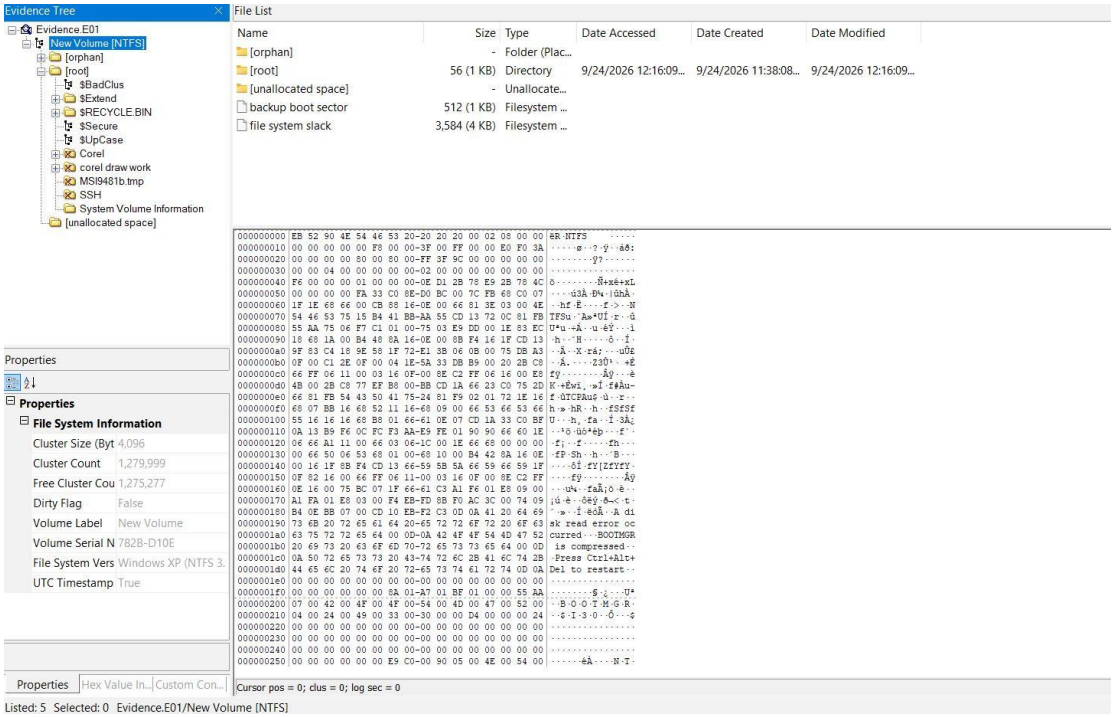
Sponsored by: Cyber Triage https://www.cybertriage.com

Product Version: Autopsy 4.23.1 (RELEASE)
Sleuth Kit Version: 4.15.0
Netbeans RCP Build: 15-387759c96ce1b891ec45ffaf524a53499455fe1a
Java: 21.0.10; Java HotSpot(TM) 64-Bit Server VM 21.0.10+8-LTS-217


About R-Drive Image
Drive Image
R-Drive Image 7.3 (Build 7314)
Copyright (c) 2001-2026 R-Tools Technology Inc.
Home site: http://www.drive-image.com/
Technical support: http://www.r-tt.com/SupportRequest.shtml
Customer service: http://www.r-tt.com/PurchasingRequest.shtml
Register...
This product is licensed to:
Licensee: UNREGISTERED
Product: R-Drive Image
License Type: Time-limited Trial
Number of computers: 1
Hardware code: Y2W6-6LR8
Days left: 24
Third Party Copyright Notice and Disclaimer
View license
Enable logging
Log file path: Enter file name
...
Write system dump...
Ok

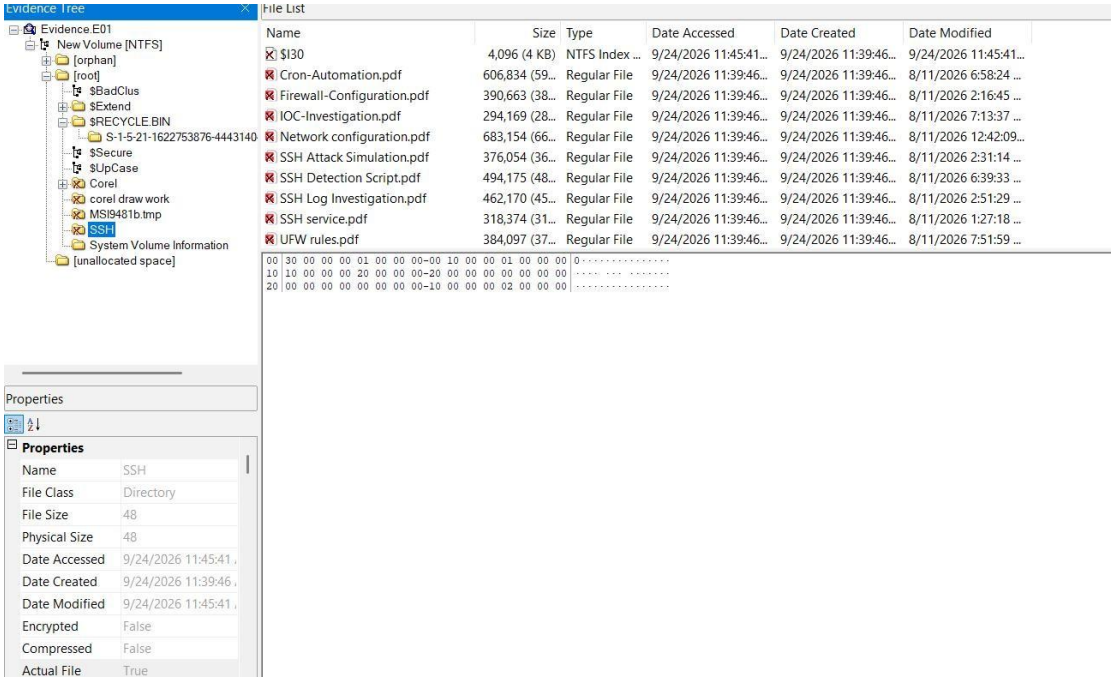

ruhuoma@ruhuoma: ~
(ruhuoma㉿ruhuoma)-[~]
$ dd --version
d (coreutils) 9.7
ackaged by Debian (9.7-3)
opyright (C) 2025 Free Software Foundation, Inc.
icense GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
his is free software: you are free to change and redistribute it.
here is NO WARRANTY, to the extent permitted by law.
ritten by Paul Rubin, David MacKenzie, and Stuart Kemp.

FTK IMAGER FINDINGS

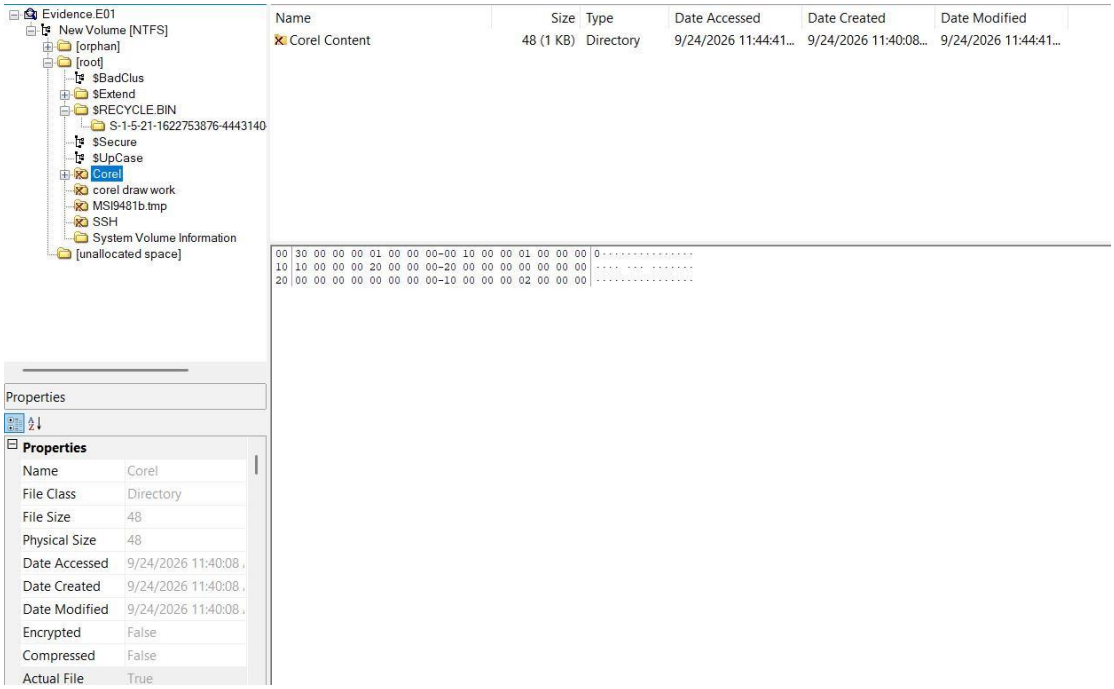


The following folders and their contents were present:

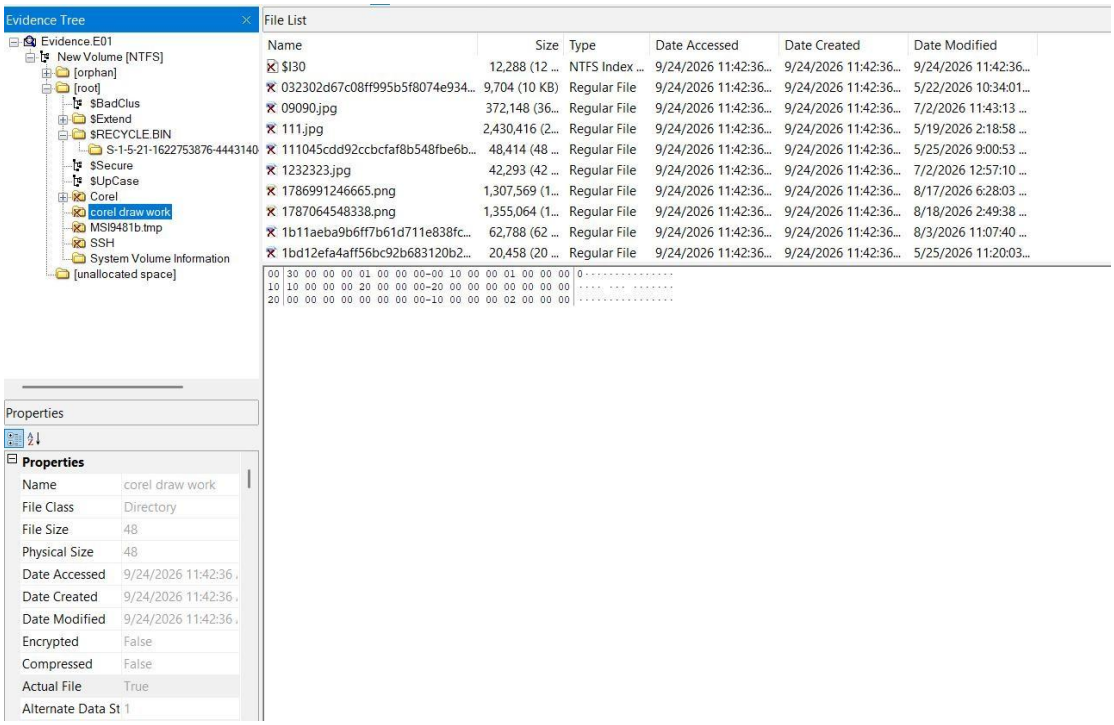
1. SSH (Containing Security-related PDF files)



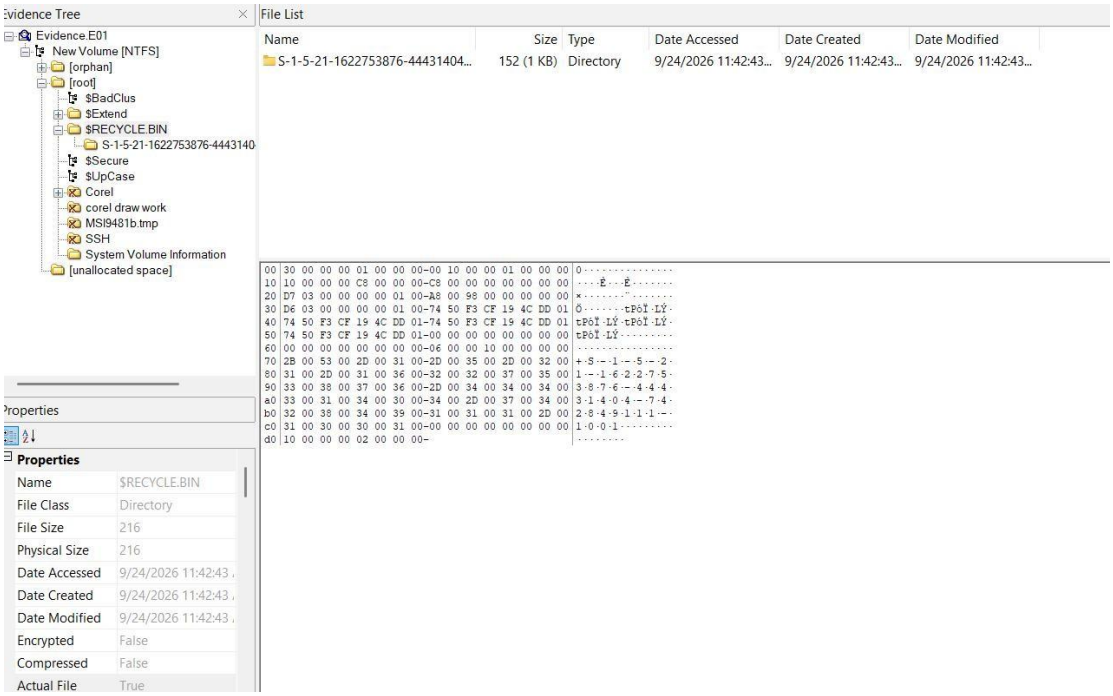
2. Corel (Containing corel content,size,type,data accessed,data created



3. Corel draw work (Containing multiple JPG and PNG image files)

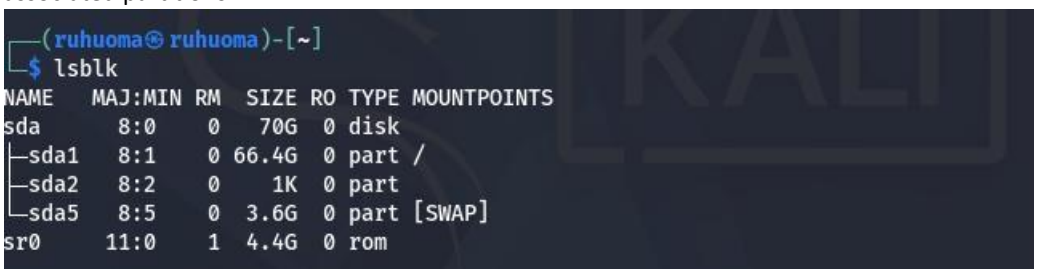


4. My pic was not in the root,recycle bin or orphan
5. Our pic was not in the root,recycle bin or orphan
6. ADDS project was not in the root,recycle bin or orphan



DD FINDINGS

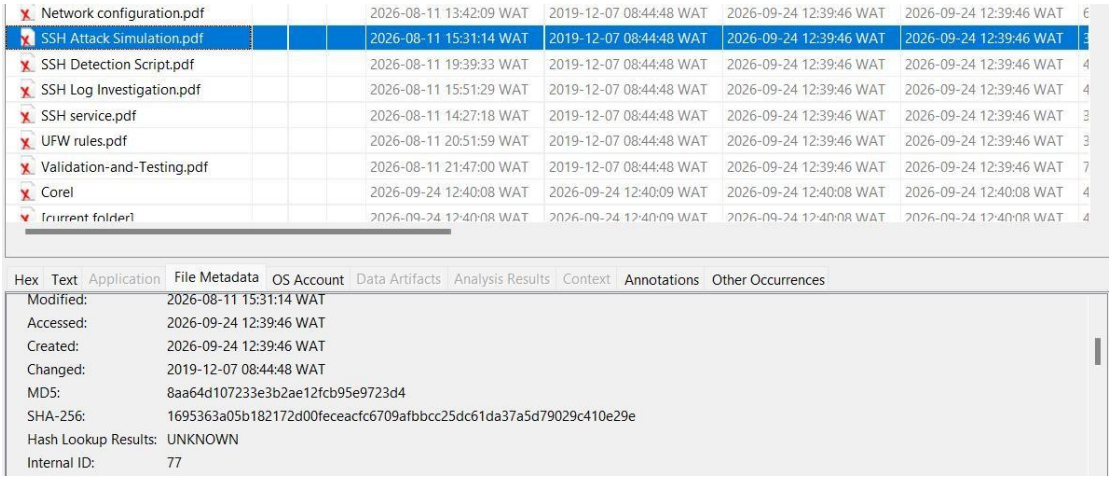
DD was used in conjunction with the Linux environment to identify the available storage devices or partitions. The lsblk command was executed to display the detected block devices and their associated partitions.

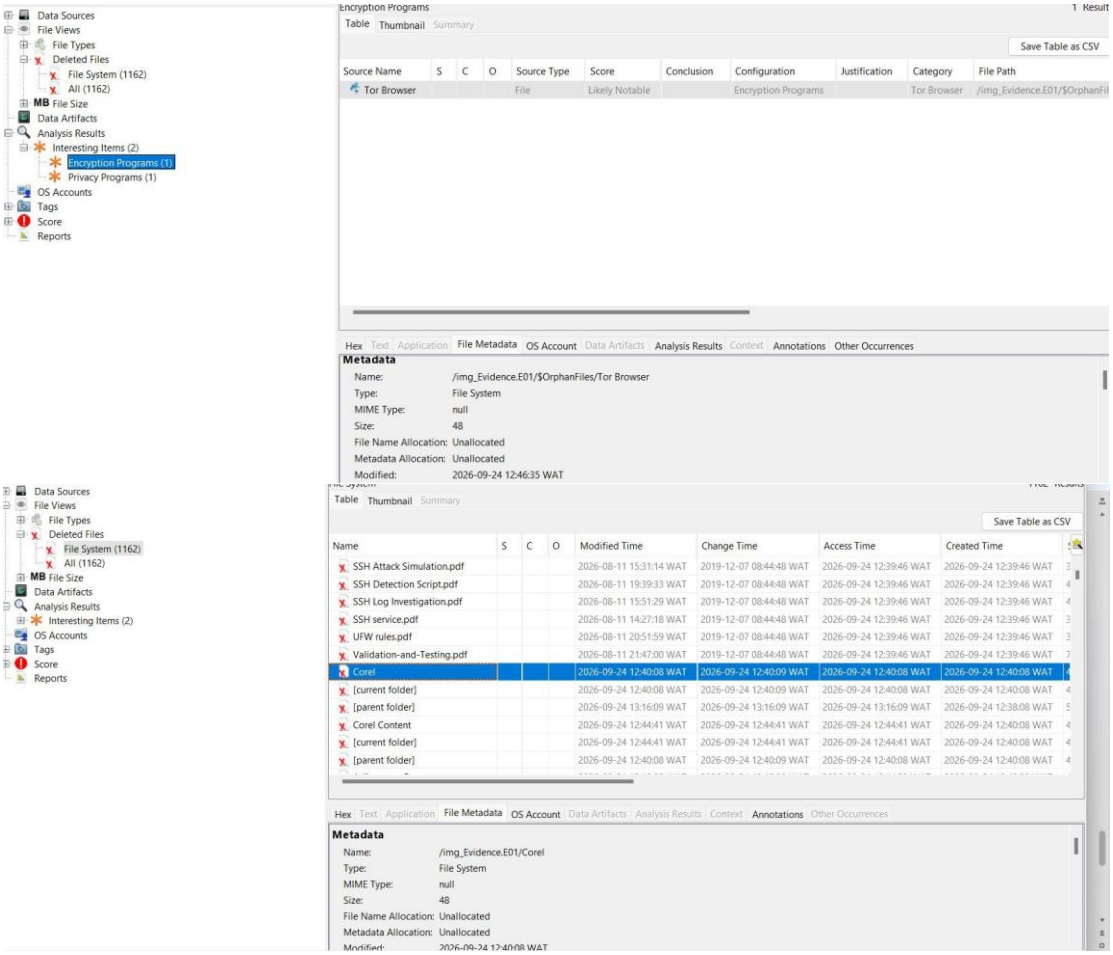


AUTOPSY FINDINGS



Autopsy Forensic Report for case All





DELETED DATA RECOVERY

The Forensic image was examined using Autopsy to identify deleted and unallocated data.The analysis identified approximately 1,162 deleted file entries within the examined file system.Several deleted

documents were identified,including files relating to SSH,corel,firewall configuration,network configuration and validation\testing. One of the examined items was ‘’Evidence01-SSH Detection Script.pdf ‘’. Autopsy identified the file as an unallocated file and provided associated metadata,timestamps,MD5 hash and SHA-256 hash values. These details were recorded as part of the forensic examination. The recovery of these deleted entries demonstrates that data that was no longer normally accessible through the file system could still be identified through forensic analysis of the acquired evidence image.

TARGET FOLDER ANALYSIS

The Forensic image was examined for relevant files and folders associated with the investigation. Particular attention was given to deleted and unallocated files identified by Autopsy,including SSHrelated documents,firewall configuration files,etc..

The examination of the selected files included reviewing their file type,size,allocation status,timestamps,MD5 AND SHA-256 hash values,and other available metadata.Where a specific filesystem path was not displayed by the forensic analysis interface,no path was inferred or assumed.

TIMELINE ANALYSIS

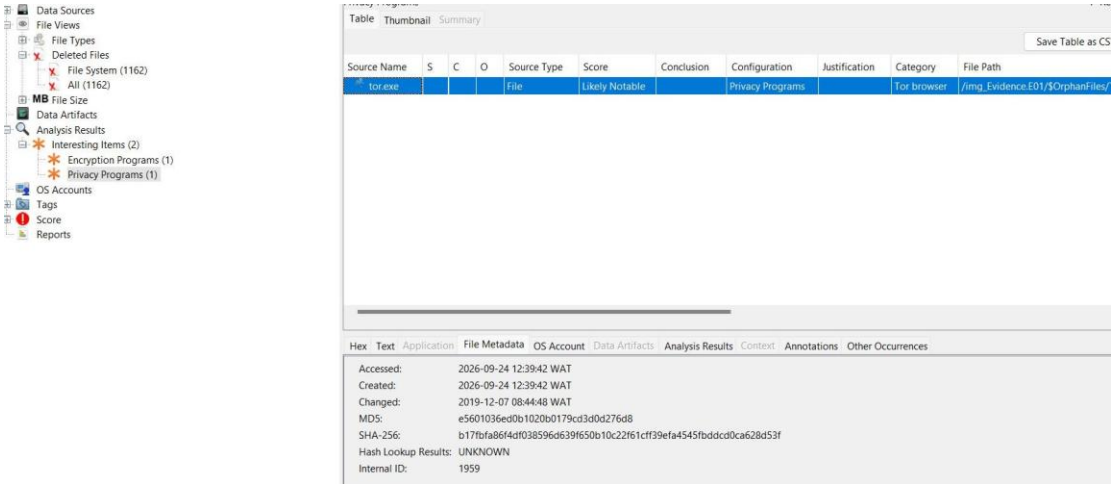
Modified:	2026-08-11 19:39:33 WAT
Accessed:	2026-09-24 12:39:46 WAT
Created:	2026-09-24 12:39:46 WAT
Changed:	2019-12-07 08:44:48 WAT

CROSS VALIDATION

Evidence obtained during the examination was reviewed across Ftk Imager,DD and Autopsy.The different examination stages were compared to improve the reliability of the recorded evidence and observations.

FINDINGS

The examination revealed 1,162 deleted-file entries. Several relevant documents were identified,including SSH,Firewall,network configuration and validation/testing files. Autopsy also recorded digital evidence associated with Tor Browser and Tor.exe.Metadata and hash information were obtained from selected files.



LIMITATIONS

R-Drive could not be successfully used. Some file paths were not displayed,and the analysis was limited to the available forensic image and recovered artifacts.

METHODOLOGY

FTK Imager and DD were used for evidence acquisition and examination. Autopsy was then used to analyze deleted files,metadata,timestamps,hashes,accounts and other relevant digital evidence.

CONCLUSION

The Investigation identified deleted files and relevant artifacts,including SSH,Firewall,Network and Tor-related evidence.

RECOMMENDATIONS

Preserve the original evidence,maintain proper chain of custody,record hash values and conduct furtjher analysis where necessary.